AURIGENE

PHARMACEUTICAL SERVICES

Niche Chemistry and Peptides Case Study-3

Cyclizations

Chemistry

- Lactam
- Stapled
- Di-sulfide
- Thioether

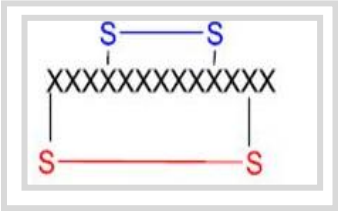
Complex cyclized Derivatives

- Head to tail
- Side chain to sidechain
- Side chain to head/tail

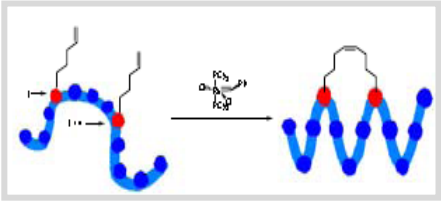
Combinations

Stabilizing extended conformation, improving potency, stability and permeability

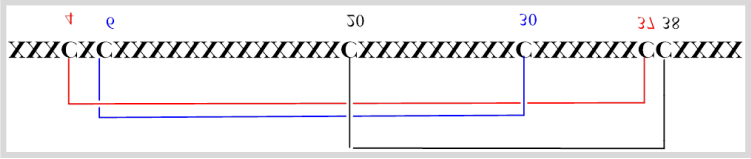
Cys-bridged peptides



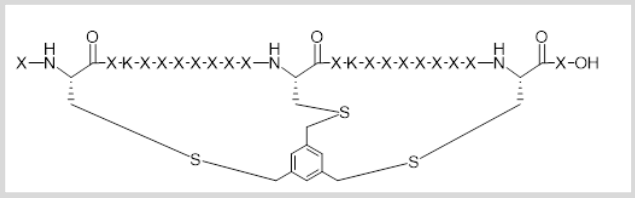
Stapled peptides



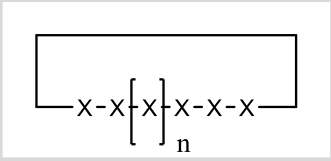
Cysteine rich peptides



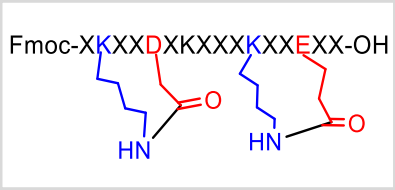
Bi-cyclic thioether linked peptides



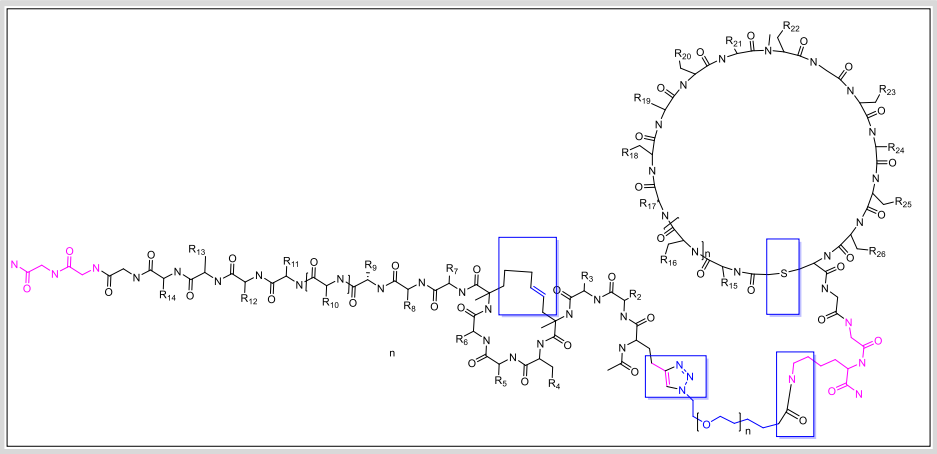
Head to tail Cyclic peptides



Lactam bridged peptides



Synthesis of complex PEG linked cyclic peptides



Conjugation/derivatization:

Chemistry

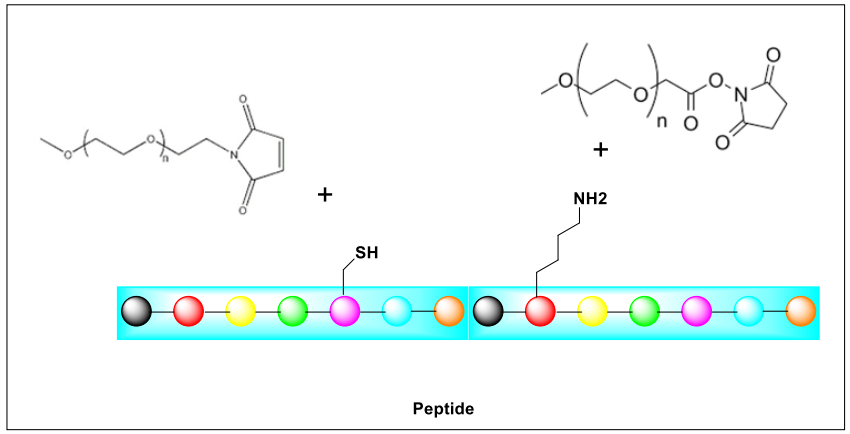
- Maleimide
- Lactam
- Click
- Ester

Derivatives

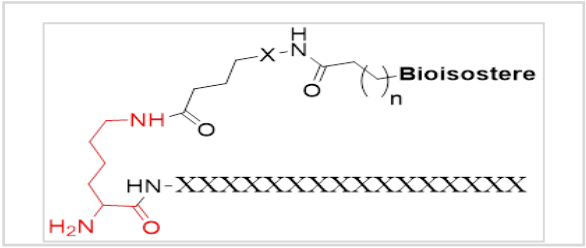
- PEG
- Lipid
- DOTA
- Oligonucleotides
- Bio-isosteres
- Florescent probes

Improving ADME parameters

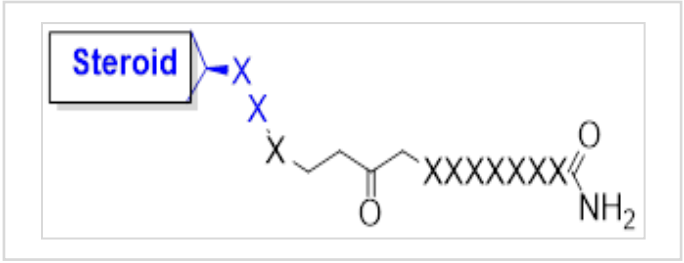
Peptide-PEG conjugate



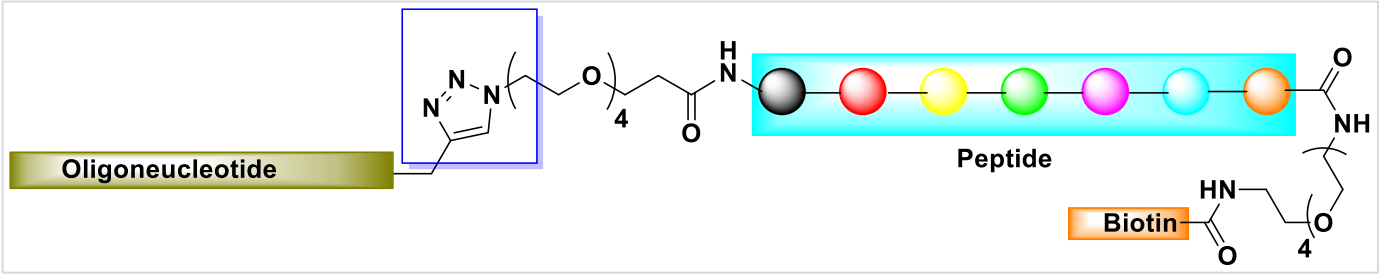
Peptides with bioisoesters



Steroid derivatisation



Peptide-oligonucleotide conjugate

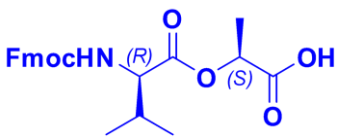


Case Studies:
Discovery

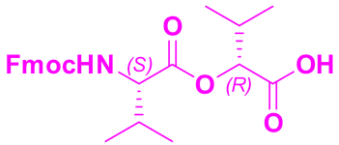
AURIGENE
PHARMACEUTICAL SERVICES

Back-bone modification and cyclization: Synthesis of cyclo-depsipeptide: Lys-Valinomycin

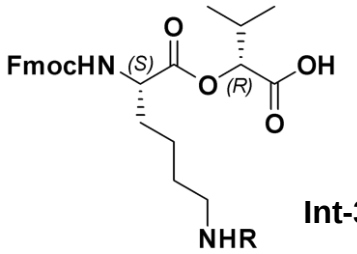
1. Synthesis of Fmoc protected depsipeptides fragments



Int-1

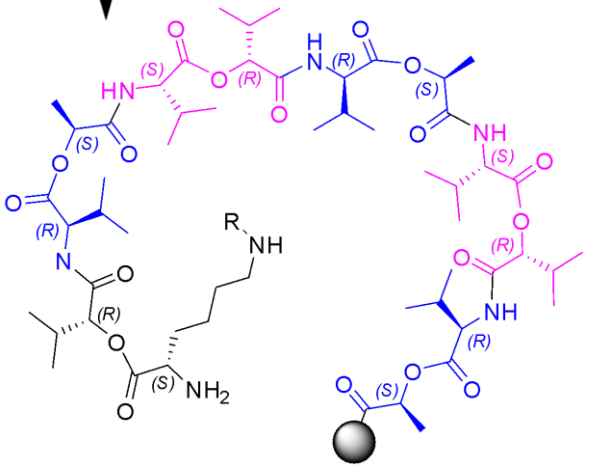


Int-2

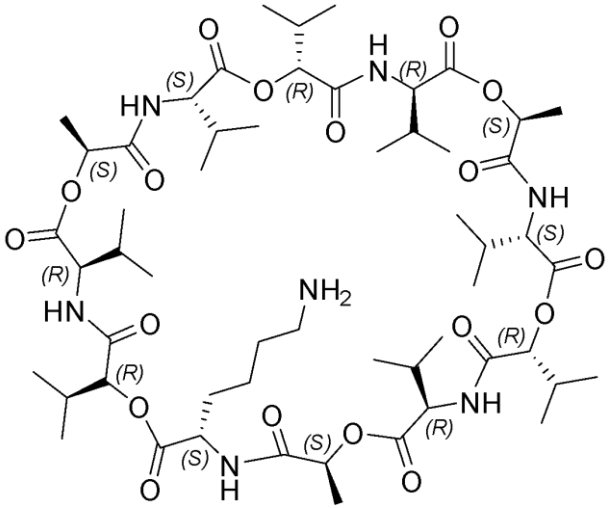


Int-3

2. SPPS with depsidipeptides

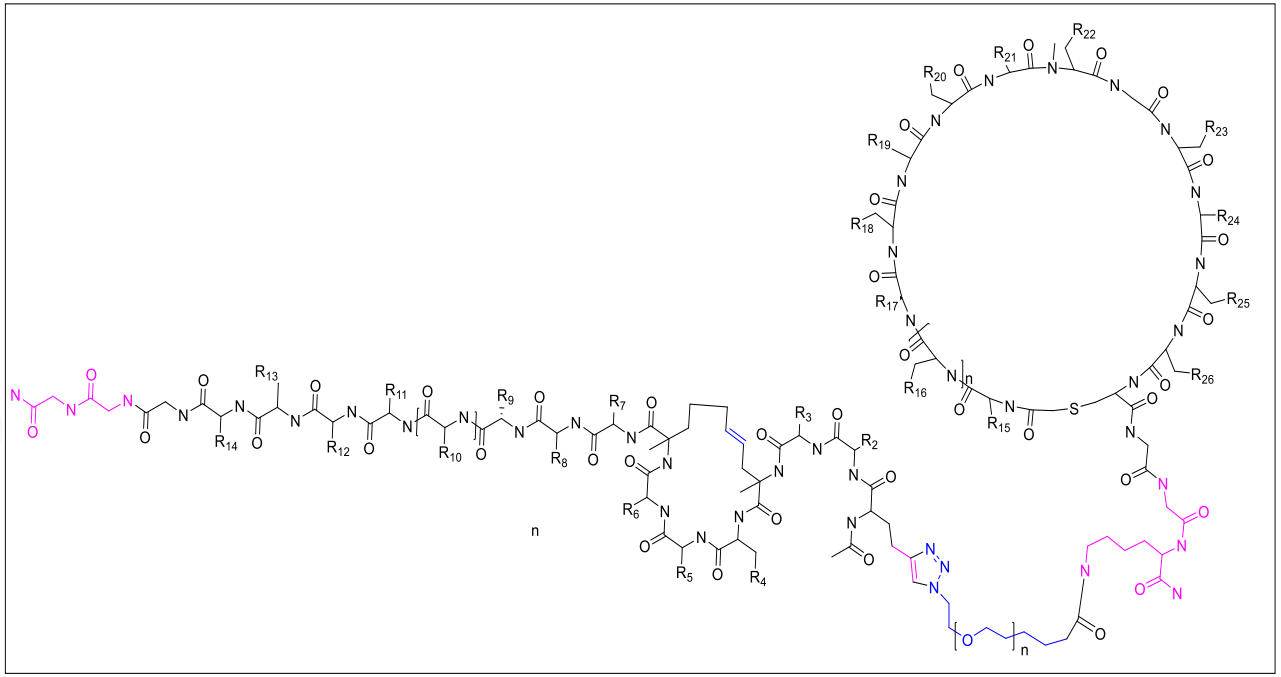


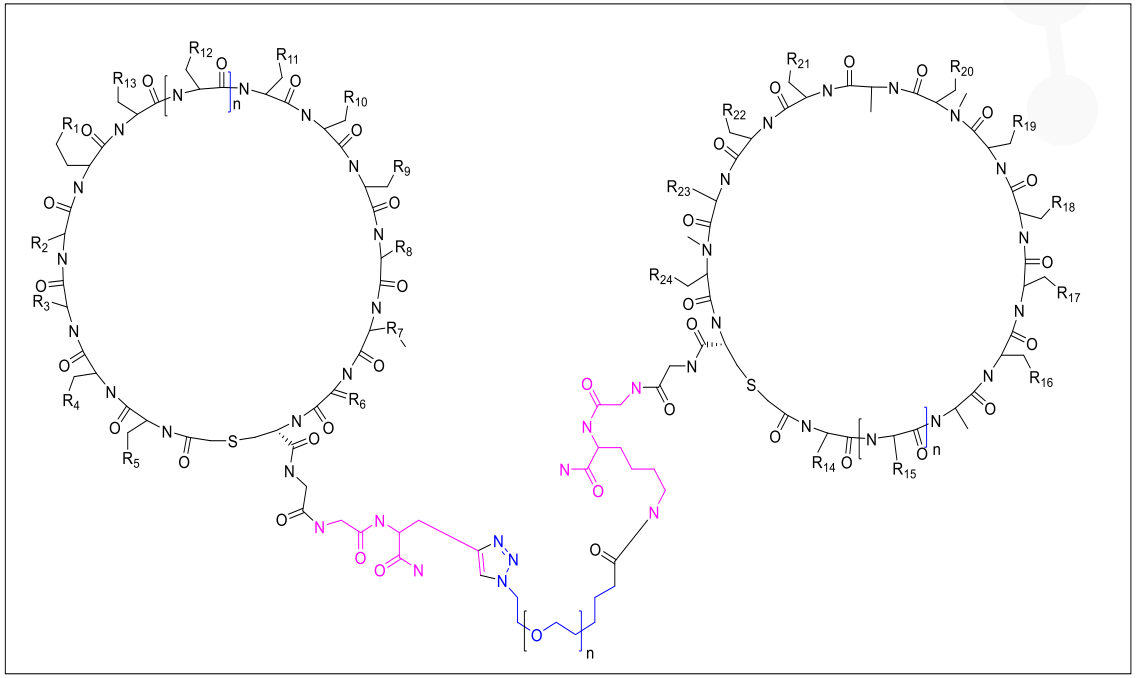
3. Cleavage and cyclization



AURIGENE
PHARMACEUTICAL SERVICES

Back-bone modification and cyclization Synthesis of complex PEG linked cyclic peptides



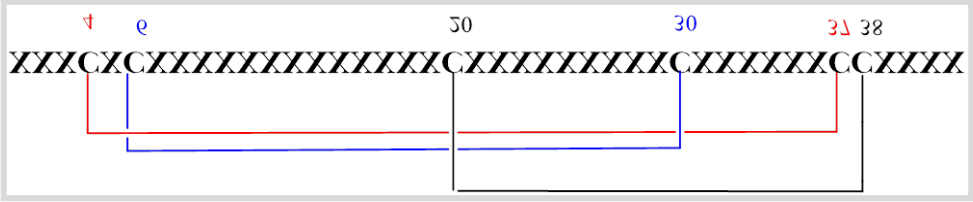


Homo and hetero functionalized PEG tethered to cyclic thioether/stapled peptides

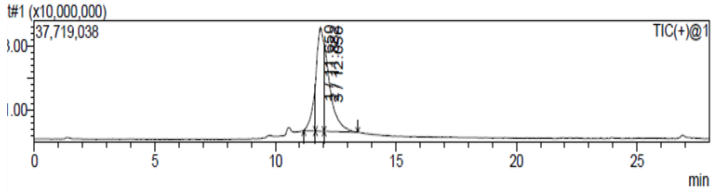
AURIGENE
PHARMACEUTICAL SERVICES

Cyclization: Synthesis and Purification of Cys-rich large peptides

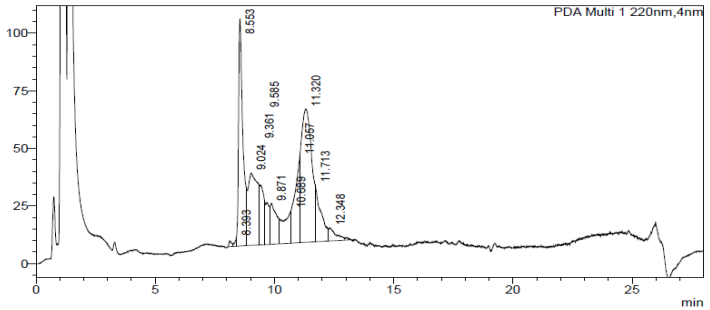
XXXXXX C(1) XXXXXXXXXXX C(2) XXXXXXXXXXXXXXXXXXXXXX C(3) C(1) XXX C(3) XXXXXXXXC(2) XXXXX



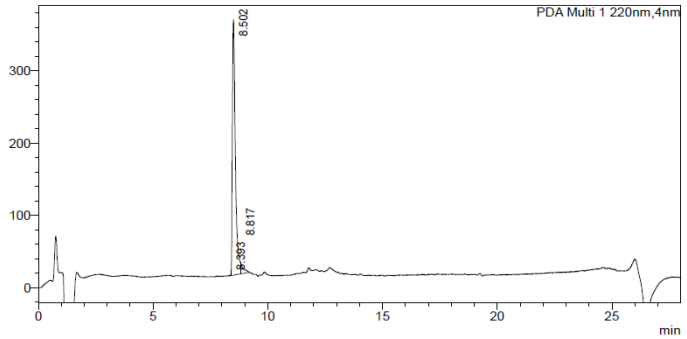
- **30 Peptides**
- **>60 amino acids**
- **25 mg-35 mg**
- **Biomimetic oxidation**



Crude peptide



Crude peptide after 24h oxidation in Glutathione redox buffer



Oxidized peptide after prep HPLC purification

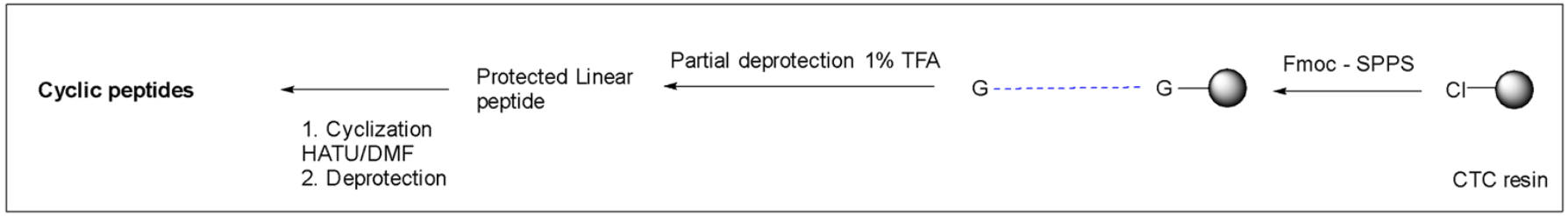
AURIGENE
PHARMACEUTICAL SERVICES

Cycilzation: Synthesis of Large head-to-tail cyclic peptides: Two different approaches

Approach 2: Head to tail cyclization of the side chain protected peptide in solution by amide coupling.

X X X X X X X X G G X X X X X X X X X X X X X X X X X X X X X X X X X X X X N C X X X X X X X X X X X X X X X X X X

Fmoc-Gly-OH as first amino acid for SPPS



Approach 2: Cyclization of purified linear thioester peptide by NCL

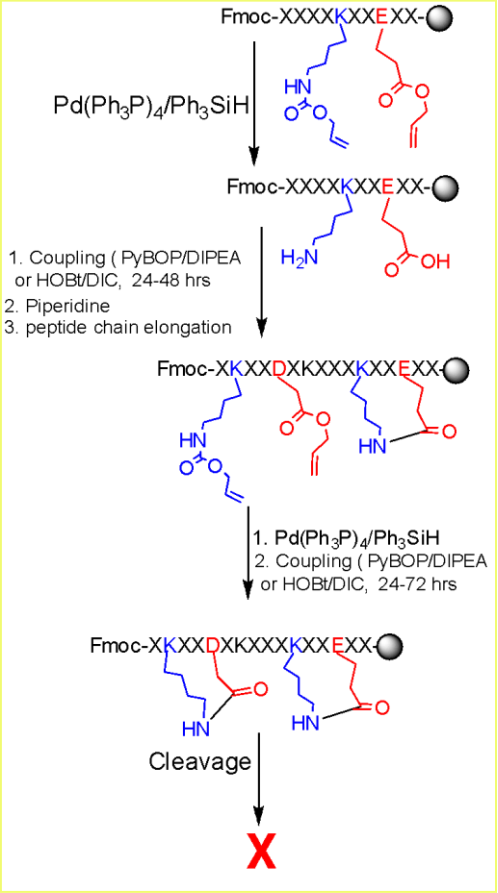
X X X X X X X X G G X X X X X X X X X X X X X X X X X X X X X X X X X X X X N C X X X X X X X X X X X X X X X X X X

Fmoc-Asp-OAllyl as first amino acid for SPPS

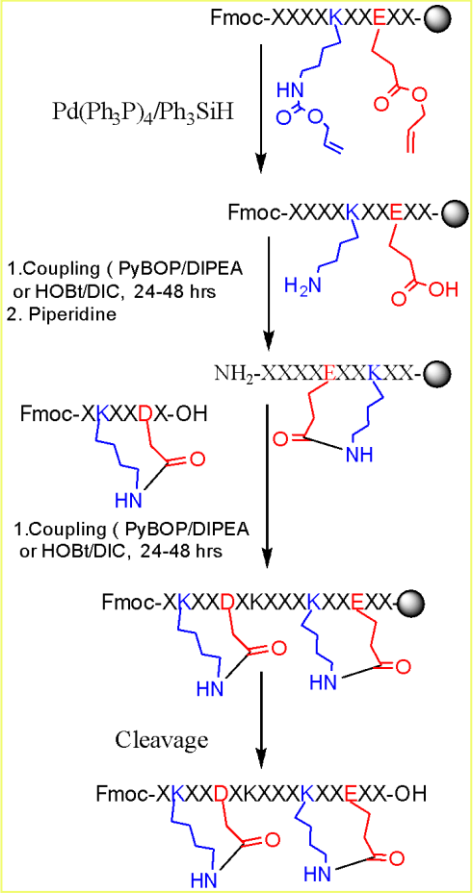
AURIGENE
PHARMACEUTICAL SERVICES

Cyclization: Double-lactam peptides

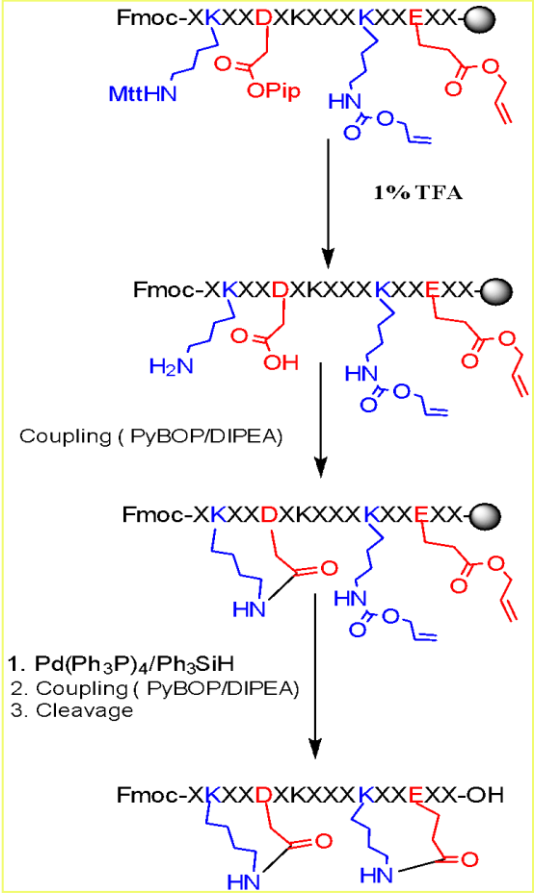
Approach 1



Approach 2



Successful Approach



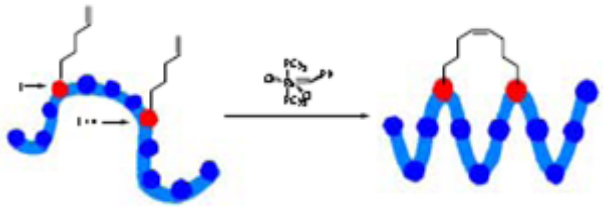
Both the approaches was not feasible to deliver the compound with required purity

Identified route for the synthesis of double lactam peptides

AURIGENE
PHARMACEUTICAL SERVICES

Cyclization: Solid Phase Synthesis of Stapled Peptides

Mono stapled



Bis-stapled

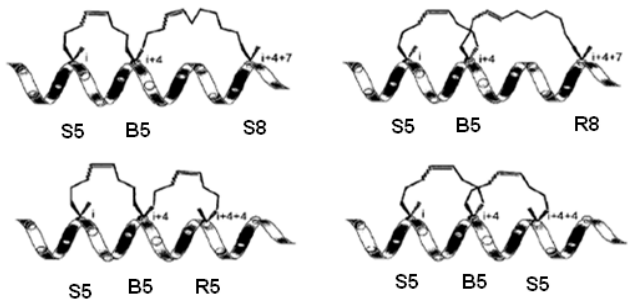
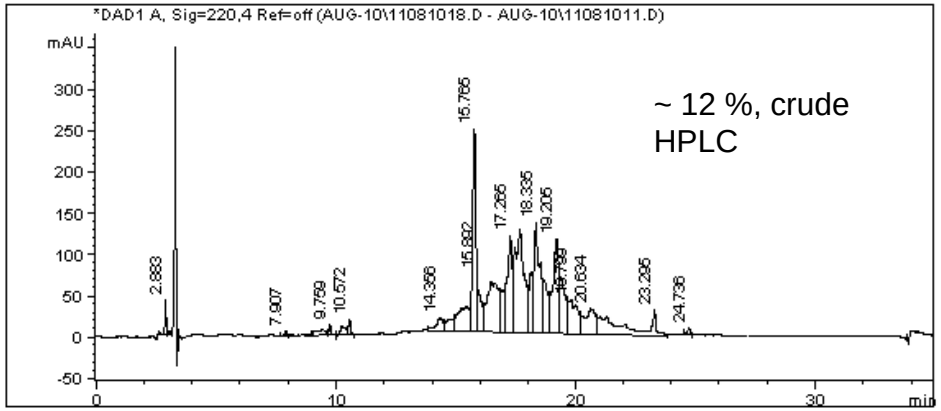


Figure adapted from WO 2008/121767

Crude HPLC of stapled peptide

AcXXXXXX[S5]XXX[B5]XXXXXX[S8]XX{Amide}

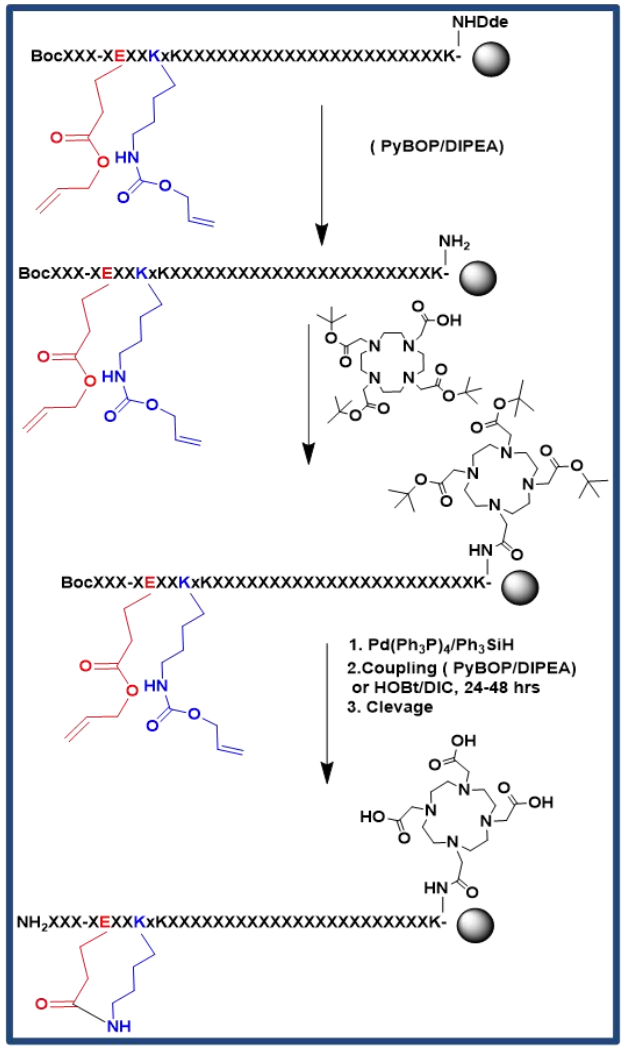


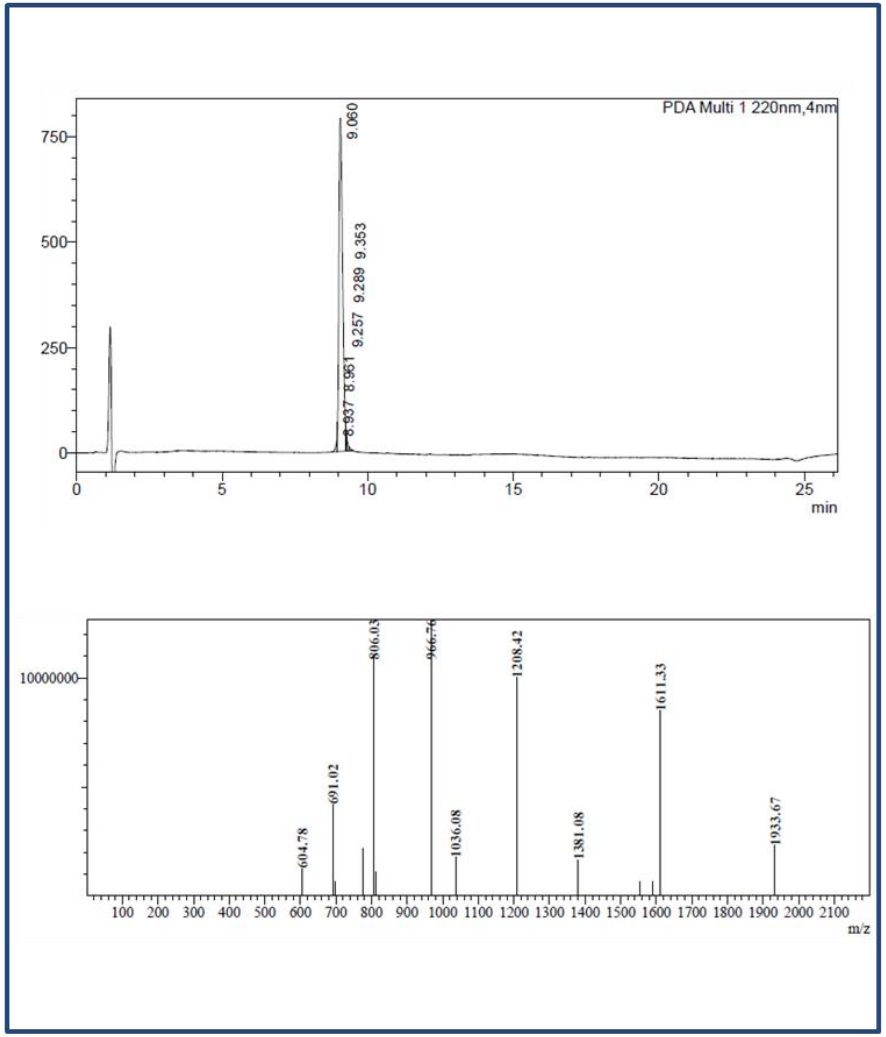
- 30 mol% Grubb's catalyst
- Classical and MW conditions
- 6 bis-stapled peptides delivered

- Stapled peptides
- Mono, Bis and reduced versions
- Up to 1000 mg

AURIGENE
PHARMACEUTICAL SERVICES

Derivatization: Peptide lactam with DOTA

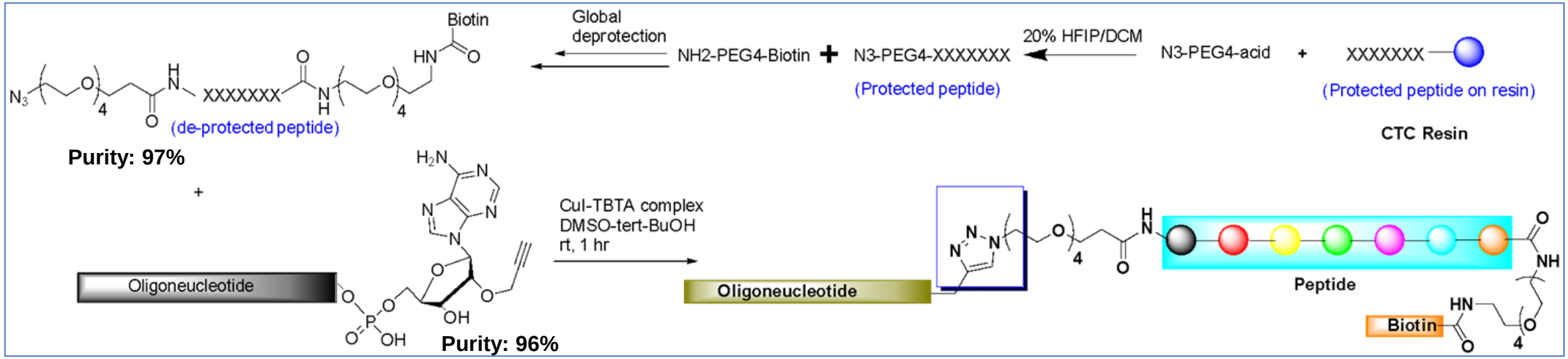




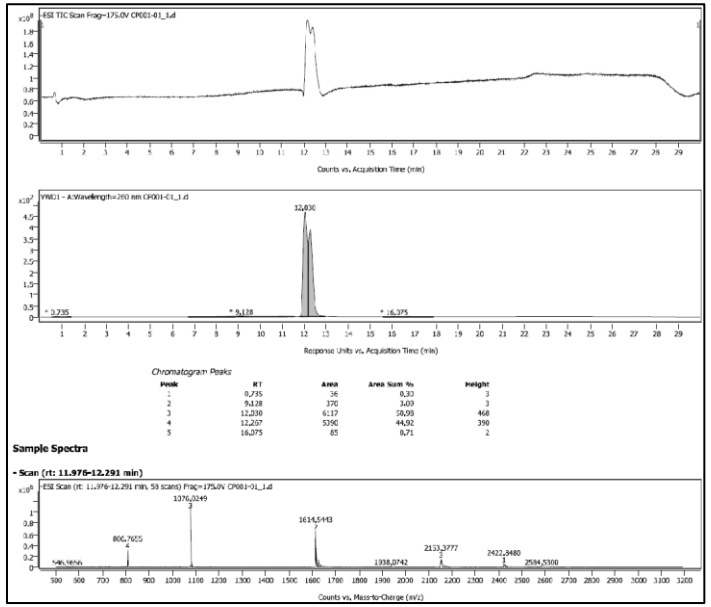
Successfully synthesized DOTA derivatized lactam peptides

AURIGENE
PHARMACEUTICAL SERVICES

Peptide-oligoneocleotide conjugate



Phosphoramidite building blocks were purchased and the oligo was assembled in-house

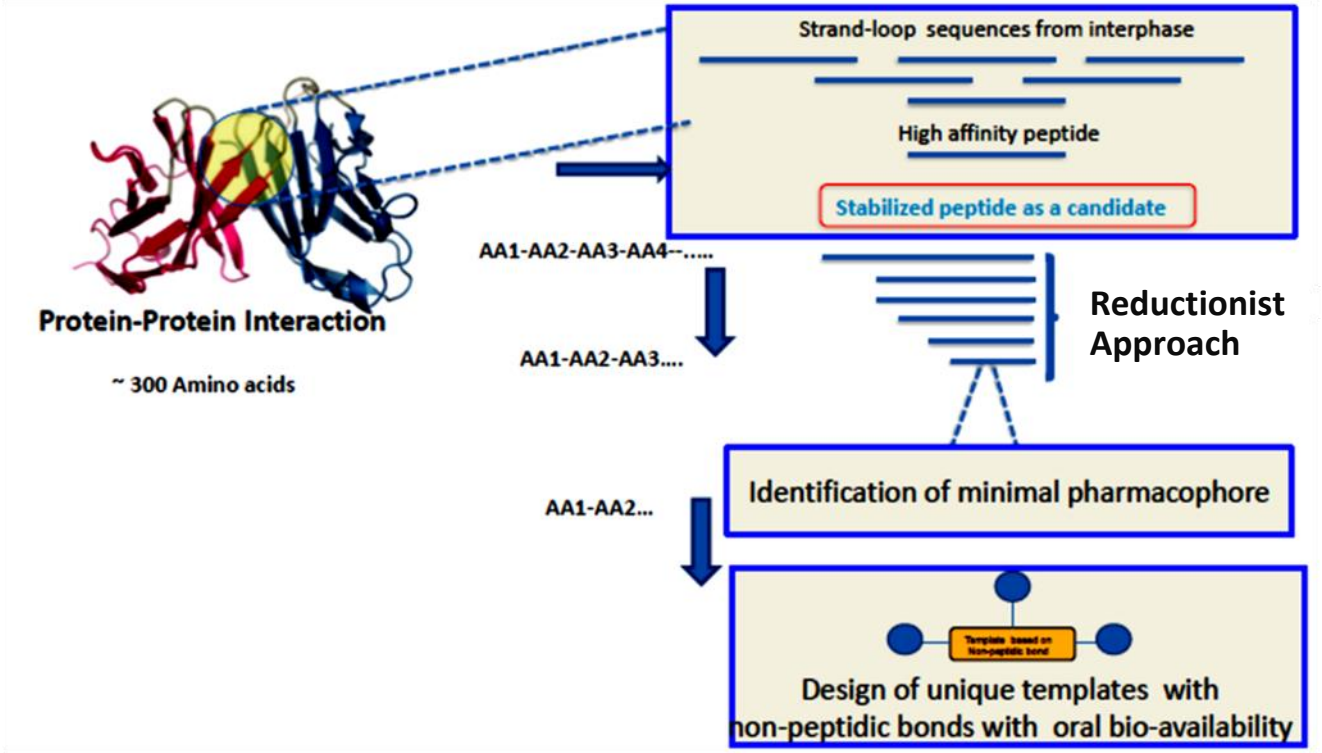


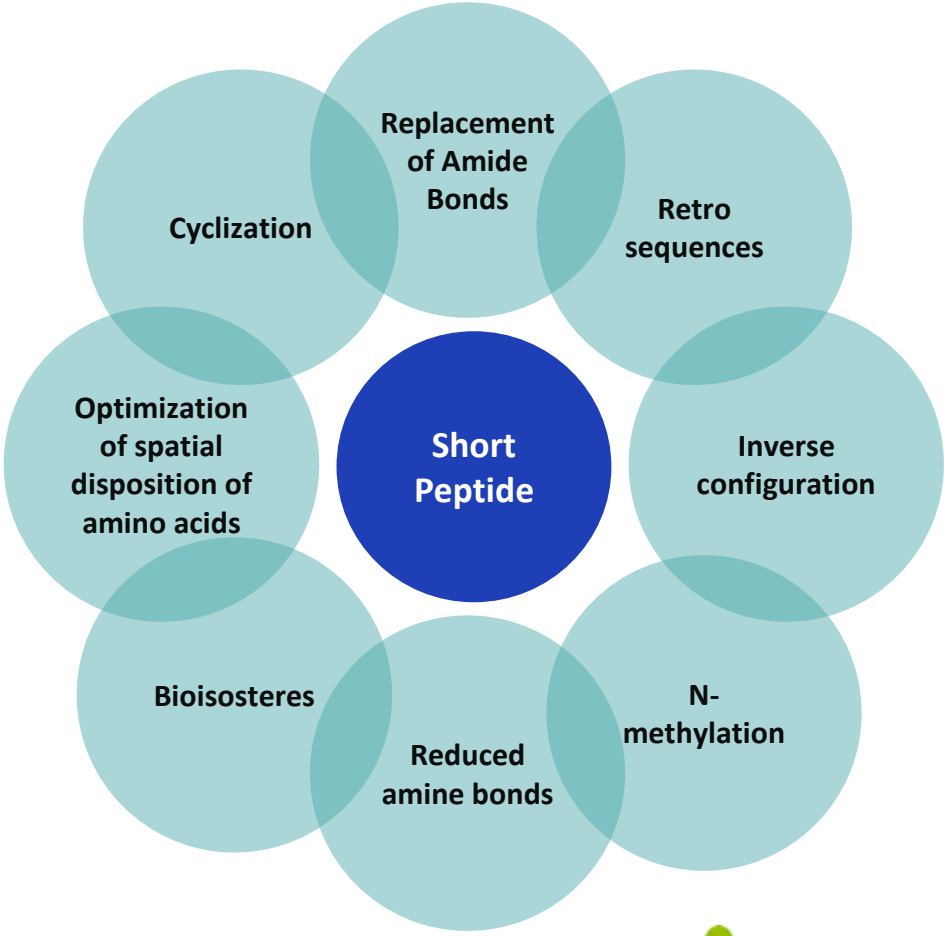
80 nMol conjugated compound was shipped

AURIGENE
PHARMACEUTICAL SERVICES

Derivatization: Converting Peptides to Peptidomimetics

Protein-Peptide-Peptidomimetics/Small molecules





Replacing peptide bonds with non peptide bonds / bioisosteres and synthesis of peptido mimetic

AURIGENE
PHARMACEUTICAL SERVICES

Case Studies:

Peptide Drug Discovery Services

AURIGENE

PHARMACEUTICAL SERVICES

Peptide Discovery Service: Metabolic Disorder

Hit Generation

- Target based IP Generation

- Synthesis, Purification & QC

- Cell based assay
- Functional activity
- Synthesis of compounds for selectivity

Lead Identification

- Plasma stability
- Solubility
- Chemical stability

- PK (i.v. & s.c.) – half life studies
- In-vitro receptor profiling - CEREP

Lead Optimization

- Bulk synthesis of lead-identified compounds
- Patent application
- In-vivo pharmacology – acute/chronic animal models
- In-vivo exploratory & In-vitro toxicology

Synthesis of Peptide Class

Peptides (Natural & Unnatural)
(up to 40 AA)
Alanine scans
Truncations

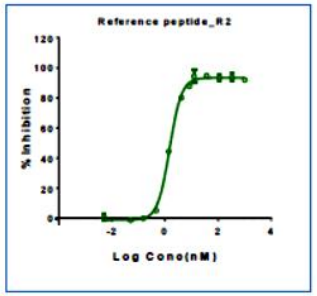
Conformationally Constrained Peptides
Lactam Stapled
Disulfide Bridged

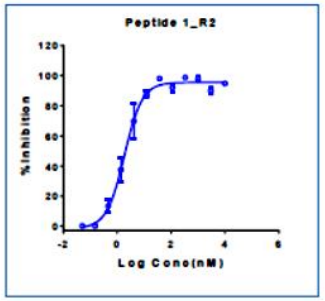
Derivatized Peptides
Lipidation, Fluorescent & dyes, PEGylations

AURIGENE
PHARMACEUTICAL SERVICES

In vitro and In Vivo Activity of Lead Peptide

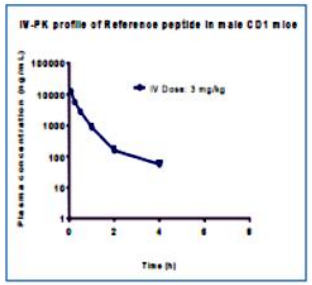
Lead Peptides: GPCR mediated activation of cAMP release in reporter cell line

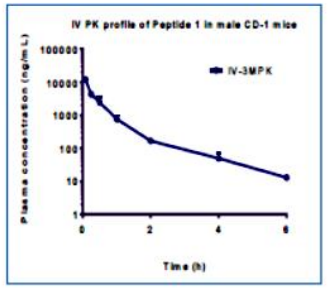




Peptide	Inhibition of receptor activation as measured by cAMP release in functional assay EC_{50} (nM)		
	R1	R2	R5
Ref peptide	158	2	76
Peptide 1	185	1.9	474

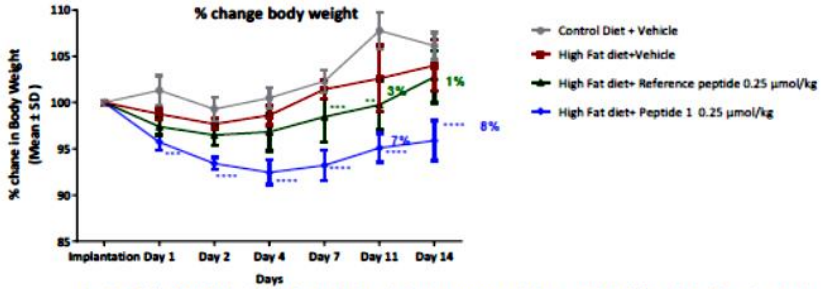
Pharmacokinetic profile of lead peptides





IV PK	Dose	$t_{1/2,\beta}$	AUC (0-last)	CL	Vd_{ss}	MRT
		(h)	(ng*h/mL)	(mL/min/Kg)	(L/kg)	(h)
Ref peptide	3 mg/kg	0.66	5456	9.08	0.25	0.46
peptide 1	3 mg/kg	1.09	4985	9.98	0.30	0.49

In vivo Efficacy: Effect of Peptide 1 and Reference peptide on the body weight in DIO mice



**p<0.01, ***p<0.001, ***p<0.0001 v/s High fat diet+vehicle when compare with Twoway ANOVA followed by Bonferroni's multiple comparisons test

- Peptide 1 showed significant reduction in body weight from day 1 onwards up to end of study (8% on day14)
- Reference peptide showed maximum efficacy of 3% reduction in body weight on day 7 and day11

AURIGENE
PHARMACEUTICAL SERVICES

© Aurigene Pharmaceutical Services Limited.

Major achievements at each targeted milestone

Hit Generation	Hit - Lead	Lead Optimization
Novel series identified	Identified selective R2 agonist with good chemical & biological stability in vitro assays	Lead peptide was well characterized in both acute and chronic *in vivo* pharmacology models
Identified potent novel R2 receptor agonists with selectivity over R1, R4 and R5 receptors in the *cAMP release functional assay*	Optimized peptide analogues to achieve favorable pharmacokinetic properties compared to that of reference peptide	Lead peptide showed significant improvement in efficacy within very well tolerated dose in mice

- **Identified highly potent, selective & patentable peptide lead for Receptor R2 agonist for treatment of obesity**
- **Lead optimized development candidate nominated from efforts at Aurigene**

AURIGENE
PHARMACEUTICAL SERVICES

© Aurigene Pharmaceutical Services Limited.